آغا خان یونیورسٹی ایگزامینیشن بورڈ

AGA KHAN UNIVERSITY EXAMINATION BOARD

Notes from E-Marking Centre on HSSC-II Business Statistics Annual Examinations 2025

Introduction

This document has been produced for the teachers and candidates of the Higher Secondary School Certificate (HSSC) Part II Business Statistics. It contains comments on candidates' responses to the 2025 HSSC-II Examination, indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses that support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding, and application skills they have developed during study. Candidates are advised to read and comprehend each question carefully before writing their response to fulfill the question's demand.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs, which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why', or 'what' may also be used. It is imperative to refer to the command word guide available on AKU-EB website for understanding the expectations of the command word.

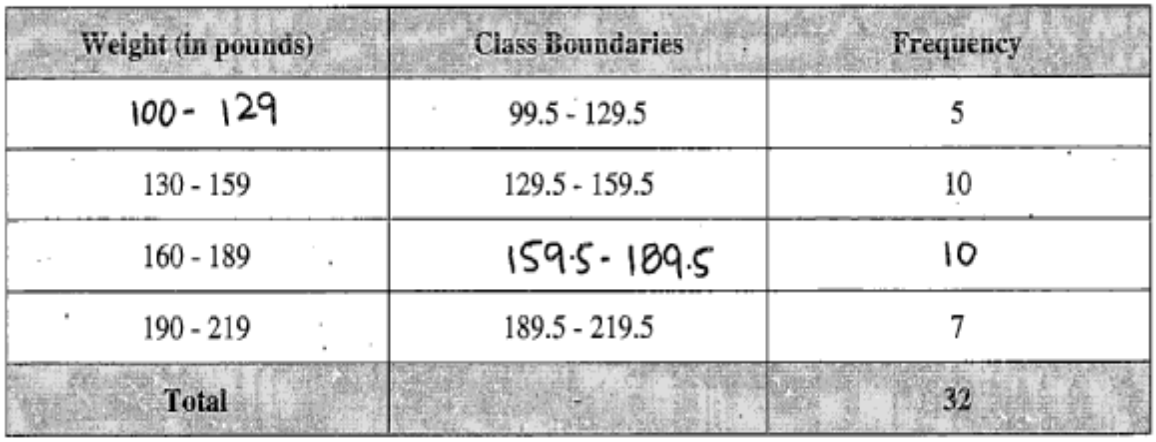
General Observations

Most candidates gave correct answers when working with **frequency distribution tables, calculating the mean of grouped data, and finding variance and standard deviation**. These topics seemed well understood.

However, many candidates found it difficult to answer questions on **quartiles, price index, permutations and combinations, and the coefficient of correlation**. Teachers should spend more time explaining these topics in class and give students extra practice so they can understand them more clearly and improve their performance.

Note: Candidates' responses shown in this report have not been corrected for grammar, spelling, format, or information.

DETAILED COMMENTS
Constructed Response Questions (CRQs)

<table>
<tr><th colspan="2">Question No. 1</th></tr>
<tr><td>Question Text</td><td>Complete the given frequency distribution table.
<table>
<tr><th>Weight (in pounds)</th><th>Class Boundaries</th><th>Frequency</th></tr>
<tr><td></td><td>99.5 - 129.5</td><td>5</td></tr>
<tr><td>130 - 159</td><td>129.5 - 159.5</td><td>10</td></tr>
<tr><td>160 - 189</td><td></td><td></td></tr>
<tr><td>190 - 219</td><td>189.5 - 219.5</td><td>7</td></tr>
<tr><td>Total</td><td>-</td><td>32</td></tr>
</table></td></tr>
<tr><td>SLO No.</td><td>10.1.1</td></tr>
<tr><td>SLO Text</td><td>Construct a frequency distribution table of discrete and continuous data including equal and unequal class intervals, tally marks/ frequency, upper class limit/ boundary, lower class limit/ boundary, class mark/ midpoint, class width and cumulative frequency.</td></tr>
<tr><td>Max Marks</td><td>3</td></tr>
<tr><td>Cognitive Level</td><td>*U</td></tr>
<tr><td>Checking Hints</td><td>1 mark for each blank (3 marks)</td></tr>
<tr><td>Overall Performance</td><td>Most of the candidates answered the question well, correctly filling in the weight ranges and class boundaries. However, some lost marks due to incorrect entries in these areas.</td></tr>
<tr><td>Description of Better Responses</td><td>Better responses represented a clear understanding of class intervals and how they align with class boundaries. Most candidates performed well by correctly filling in the missing values. They accurately identified the weight range of 100–129 for the first row, matching it with the given class boundaries of 99.5–129.5. For the class boundaries, many correctly entered 159.5–189.5 to correspond with the weight range of 160–189. They also maintained the correct frequency of 10 for that class and ensured that the total frequency remained consistent at 32. These responses demonstrated accuracy, attention to detail, and a solid grasp of how grouped data should be presented.</td></tr>
<tr><td>Image of Better Response</td><td> 
<table>
<tr><th>Weight (in pounds)</th><th>Class Boundaries</th><th>Frequency</th></tr>
<tr><td>100 - 129</td><td>99.5 - 129.5</td><td>5</td></tr>
<tr><td>130 - 159</td><td>129.5 - 159.5</td><td>10</td></tr>
<tr><td>160 - 189</td><td>159.5 - 189.5</td><td>10</td></tr>
<tr><td>190 - 219</td><td>189.5 - 219.5</td><td>7</td></tr>
<tr><td>Total</td><td>-</td><td>32</td></tr>
</table></td></tr>
<tr><td>Description of Weaker Responses</td><td>Weaker responses represented confusion in interpreting class intervals and their corresponding boundaries. Many candidates made errors in filling in the table, such as incorrectly entering the weight ranges and class boundaries. Although the frequency was often recorded correctly, mistakes in the other columns led to inconsistencies. These inaccuracies suggest a lack of understanding of how class intervals align with class boundaries, which affected the overall accuracy of the data.</td></tr>
</table>

Images of Weaker Responses

Image (i)

Weight (in pounds)	Class Boundaries	Frequency
100-5 - 130	99.5 - 129.5	5
130 - 159	129.5 - 159.5	10
160 - 189	160·5 - 189·5	10
190 - 219	189.5 - 219.5	7
Total	-	**32**

Image (ii)

Weight (in pounds)	Class Boundaries	Frequency
120 - 139	99.5 - 129.5	5
~~130 - 159~~ 5	129.5 - 159.5	10
160 - 189	155.5 - 194.5	10
190 - 219	189.5 - 219.5	7
Total	-	**32**

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	**Preferred Pedagogy** Used for this SLO**	**Assessment Strategies**
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration ** For description of each Pedagogy, refer to Annexure A	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

By implementing these strategies, educators can enhance students' understanding and application of the concepts:

Simplify Core Concepts:

Use real-life analogies (e.g., class intervals like age groups in a survey).

Differentiate discrete (counted) vs. continuous (measured) data with clear examples.

Address Common Mistakes:

Misaligned Boundaries: Stress the "±0.5 adjustment" for continuous data.

Incorrect Class Widths: Use a formula (Width = Upper Bound – Lower Bound).

Interactive Learning:

Have students collect and tabulate classroom data (e.g., height, age, marks etc).

Use Excel/ Google Sheets to auto-generate tables for verification.

Reinforce with Visuals & Mnemonics:

Posters/ Flowcharts: Summarise steps for quick reference.

Progressive Difficulty:
Start with small, equal intervals (e.g., test scores).
Advance to unequal intervals (e.g., income ranges) and real-world datasets.

*K = Knowledge U = Understanding A = Application and other higher-order cognitive skills

Question No. 2a

Candidates were given the choice to attempt any ONE out of the two questions: 2a and 2b.

Question Text

In a sports academy, players of different age groups are registered. These age groups are recorded in the given table. Complete the frequency table to find the average age of the players.

Age Groups	Number of Players (*f*)		
14 - 17	10		
18 - 21	25		
22 - 25	30		
26 - 29	15		
30 - 33	20		
34 - 37	14		

SLO No. 11.1.1

SLO Text Calculate arithmetic mean (A.M) for grouped and ungrouped data.

Max Marks 3

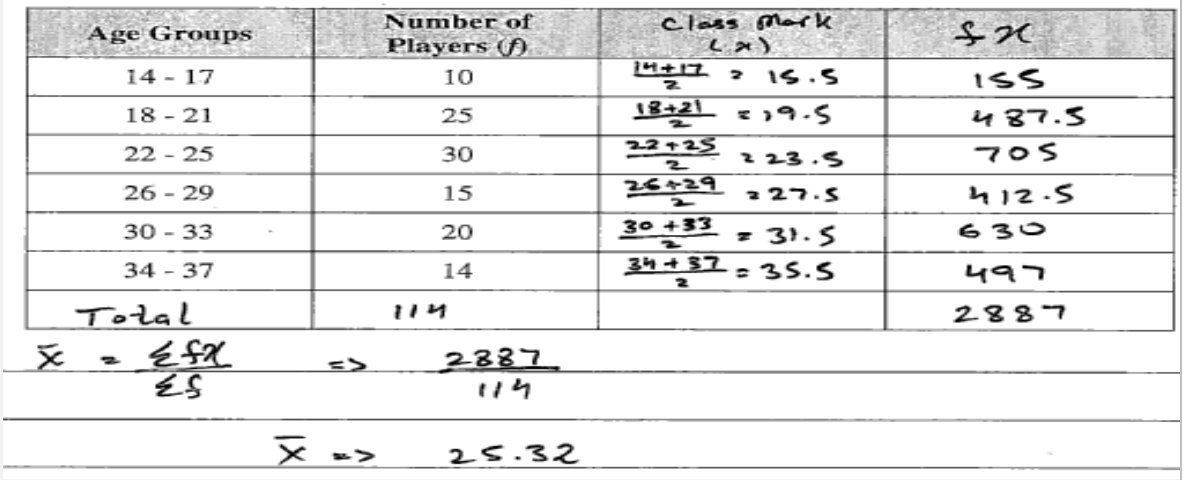
Cognitive Level U

Checking Hints
1 mark for each column (2 required)
1 mark for calculating the mean by using the formula

Overall Performance
This question was attempted by the majority of candidates compared to part (b). Most showed a clear understanding of the concept, using correct methods and providing accurate step-by-step calculations. Their work was generally well organised. However, some candidates showed confusion with basic concepts, such as using incorrect formulas and misinterpreting class boundaries and limits. These errors often led to incomplete or incorrect solutions, indicating the need for further clarification and practice on these topics.

Description of Better Responses
Better responses represented a clear understanding of the required calculations. Candidates accurately found the midpoints of each class, correctly calculated the *fx* values, and determined the mean using the appropriate formula. Their solutions were well-organised and followed a logical step-by-step process, reflecting a strong grasp of the method and its application.

Image of Better Response



Age Groups	Number of Players (*f*)	Class Mark (*x*)	*fx*
14 - 17	10	$\frac{14+17}{2} = 15.5$	155
18 - 21	25	$\frac{18+21}{2} = 19.5$	487.5
22 - 25	30	$\frac{22+25}{2} = 23.5$	705
26 - 29	15	$\frac{26+29}{2} = 27.5$	412.5
30 - 33	20	$\frac{30+33}{2} = 31.5$	630
34 - 37	14	$\frac{34+37}{2} = 35.5$	497
Total	114		2887

$\bar{x} = \frac{\Sigma fx}{\Sigma f} \Rightarrow \frac{2887}{114}$

$\bar{x} \Rightarrow 25.32$

Description of Weaker Responses
Weaker responses exhibited several fundamental errors that affected the accuracy of the solutions. Common mistakes included confusing class boundaries with class limits, calculating cumulative frequency instead of using the given frequency, and applying

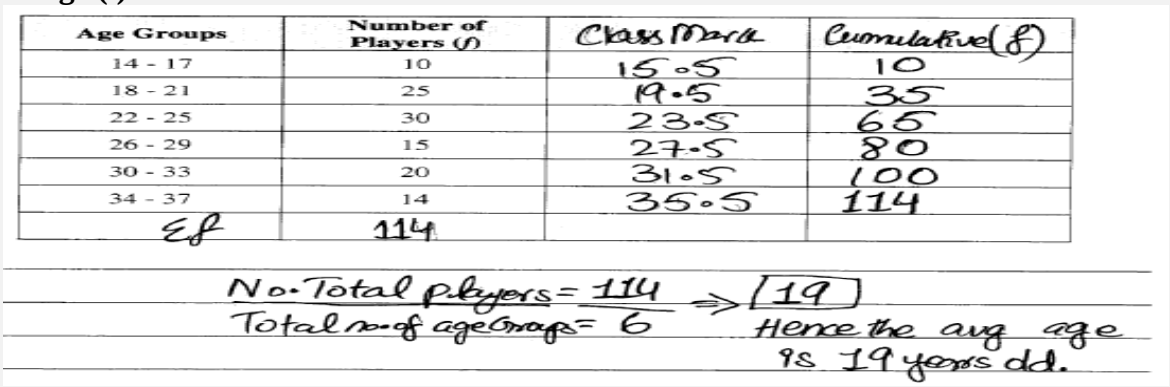
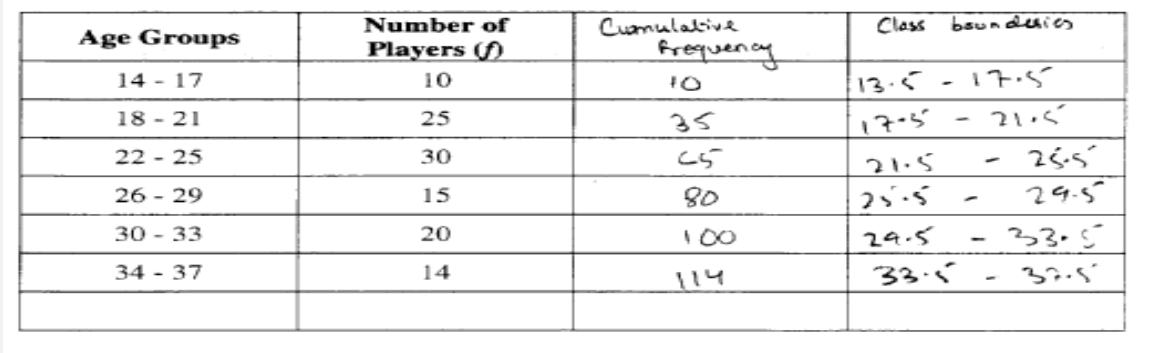
	incorrect formulas for finding the mean. In many cases, solutions were left incomplete, indicating a lack of clarity in the procedural steps. These misunderstandings and procedural errors consistently resulted in incorrect answers and loss of marks.
Images of Weaker Response	**Image (i)**  **Image (ii)** 

Image (i) transcription:

Age Groups	Number of Players (f)	Class Mark	Cumulative (f)
14 - 17	10	15.5	10
18 - 21	25	19.5	35
22 - 25	30	23.5	65
26 - 29	15	27.5	80
30 - 33	20	31.5	100
34 - 37	14	35.5	114
Ef	114		

No. Total players = 114 / Total no. of age Groups = 6 ⇒ [19]
Hence the avg age is 19 years old.

Image (ii) transcription:

Age Groups	Number of Players (f)	Cumulative Frequency	Class boundaries
14 - 17	10	10	13.5 - 17.5
18 - 21	25	35	17.5 - 21.5
22 - 25	30	65	21.5 - 25.5
26 - 29	15	80	25.5 - 29.5
30 - 33	20	100	29.5 - 33.5
34 - 37	14	114	33.5 - 37.5

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:
To enhance candidates' performance in this concept, teachers may consider the following strategies:
Concrete Examples:
Start with daily-life scenarios (e.g., "Find your average screen time over the past 5 days").
Hands-on Activities:
Have students collect their data (e.g., number of siblings of classmates, shoe sizes, etc.)
Then use it to compute the mean.
Use Visual Aids:
Number lines and bar charts to show distribution and the balance point.

Question No. 2b

Candidates were given the choice to attempt any ONE out of the two questions: 2a and 2b.

Question Text

Consider the given frequency distribution.

Class intervals	Class boundaries	Frequency	Cumulative Frequency
0 – 9	0.5 - 9.5	5	5
10 – 19	9.5 - 19.5	7	12
20 – 29	19.5 - 29.5	12	24
30 – 39	29.5 - 39.5	10	34
40 – 49	39.5 - 49.5	8	42
50 – 59	49.5 - 59.5	13	55
60 – 69	59.5 - 69.5	10	65

Using the given frequency distribution, calculate the

i. 1st quartile.

ii. 3rd quartile.

SLO No. 11.3.2

SLO Text Calculate quartiles for ungrouped and grouped data.

Max Marks 3

Cognitive Level U

Checking Hints

i. 1 marks for using correct mathematical construct for quartiles
1 mark for finding the first quartile

ii. 1 mark for finding the third quartile

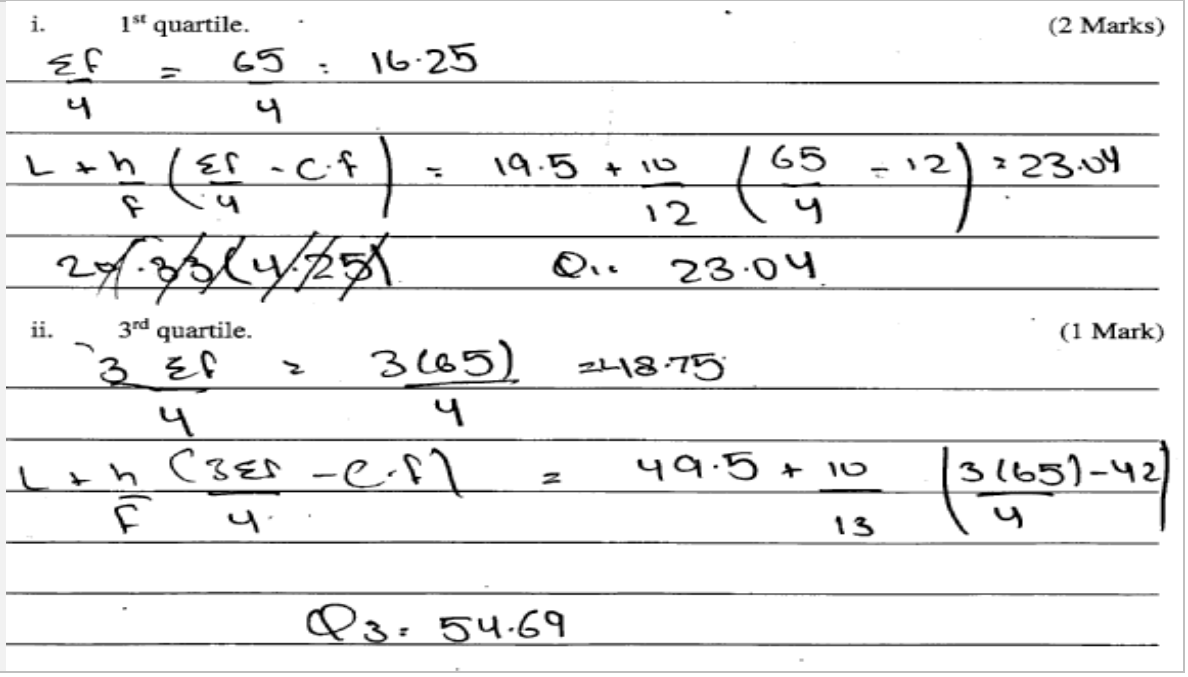
Overall Performance

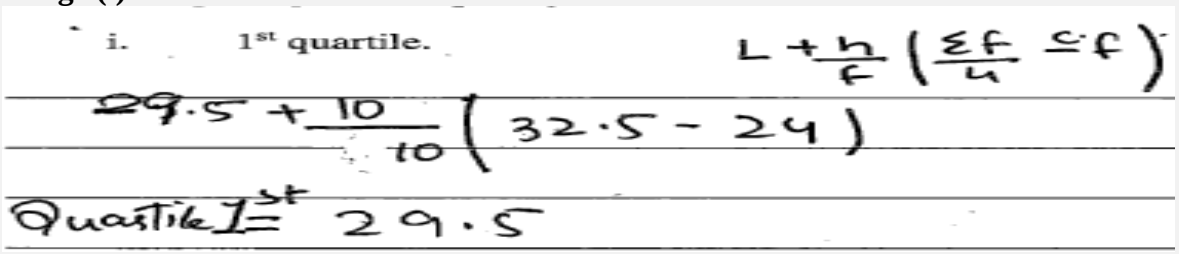
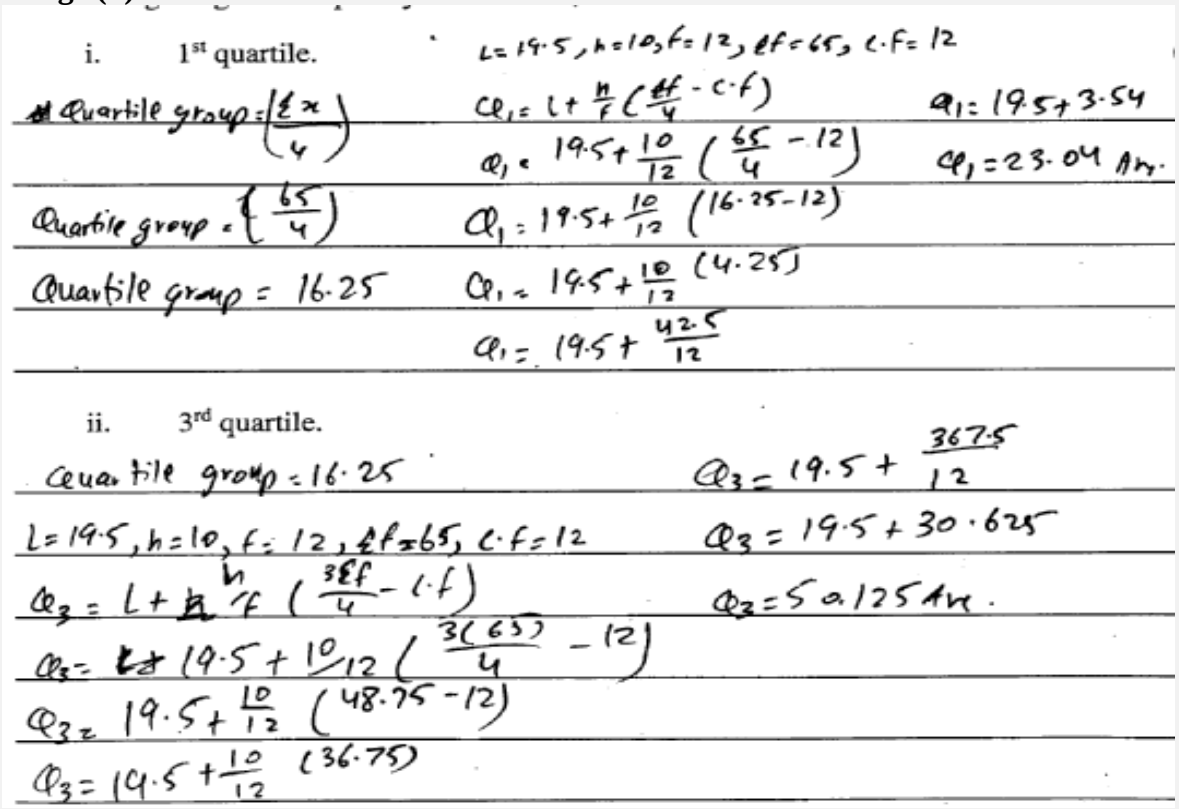
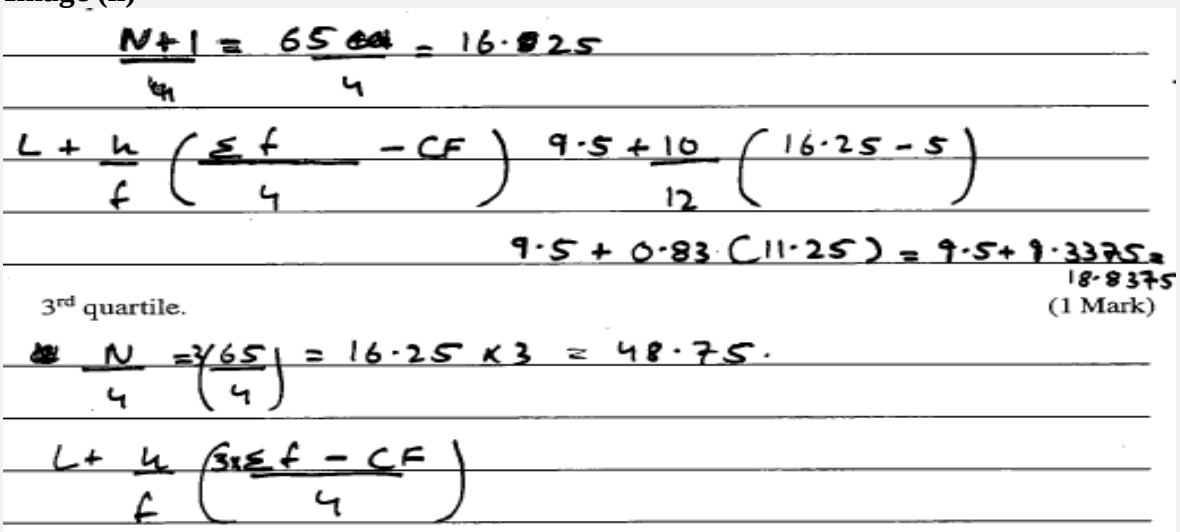
As this part of the question was optional and longer than part (a), most students chose to attempt part (a) instead, which likely explains the lower number of responses to this section. Only a few candidates attempted this part, possibly due to its length or perceived complexity. Among those who did, better responses demonstrated a clear understanding of the relevant formula and followed the correct steps to solve the problem accurately. In contrast, weaker responses showed confusion, with frequent procedural errors and incorrect application of the method. This highlights the need for more targeted practice and clear explanations to strengthen students' understanding of the required procedures.

Description of Better Responses

Better responses used the correct formula of median and followed the appropriate substitution to calculate both quartiles accurately. These candidates showed a good understanding of the method and were able to apply it correctly.

Image of Better Response



Description of Weaker Responses	Few of the candidates were able to correctly calculate the first quartile, showing a good understanding of the initial steps. However, a number of them substituted incorrect class values when attempting to find the third quartile. Some candidates used the correct class values but made errors in their calculations, which affected the final answer. In several cases, the third quartile was not attempted at all, suggesting a lack of confidence or understanding of the full procedure.
Images of Weaker Responses	**Image (i)**  i. 1st quartile. $L + \frac{h}{f}\left(\frac{\Sigma f}{4} - c.f\right)$ $29.5 + \frac{10}{10}(32.5 - 24)$ Quartile 1st = 29.5 **Image (ii)**  i. 1st quartile. $L = 19.5, h = 10, f = 12, \Sigma f = 65, c.f = 12$ Quartile group = $\left(\frac{\Sigma x}{4}\right)$ Quartile group = $\left(\frac{65}{4}\right)$ Quartile group = 16.25 $Q_1 = L + \frac{h}{f}\left(\frac{\Sigma f}{4} - c.f\right)$ $Q_1 = 19.5 + \frac{10}{12}\left(\frac{65}{4} - 12\right)$ $Q_1 = 19.5 + \frac{10}{12}(16.25 - 12)$ $Q_1 = 19.5 + \frac{10}{12}(4.25)$ $Q_1 = 19.5 + \frac{42.5}{12}$ $Q_1 = 19.5 + 3.54$ $Q_1 = 23.04$ Ans. ii. 3rd quartile. Quartile group = 16.25 $L = 19.5, h = 10, f = 12, \Sigma f = 65, c.f = 12$ $Q_3 = L + \frac{h}{f}\left(\frac{3\Sigma f}{4} - c.f\right)$ $Q_3 = 19.5 + \frac{10}{12}\left(\frac{3(65)}{4} - 12\right)$ $Q_3 = 19.5 + \frac{10}{12}(48.75 - 12)$ $Q_3 = 19.5 + \frac{10}{12}(36.75)$ $Q_3 = 19.5 + \frac{367.5}{12}$ $Q_3 = 19.5 + 30.625$ $Q_3 = 50.125$ Ans. **Image (ii)**  $\frac{N+1}{4} = \frac{65}{4} = 16.25$ $L + \frac{h}{f}\left(\frac{\Sigma f}{4} - CF\right)$ $9.5 + \frac{10}{12}(16.25 - 5)$ $9.5 + 0.83(11.25) = 9.5 + 9.3375 = 18.8375$ 3rd quartile. (1 Mark) $\frac{N}{4} = \left(\frac{65}{4}\right) = 16.25 \times 3 = 48.75.$ $L + \frac{h}{f}\left(\frac{3\Sigma f}{4} - CF\right)$

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Pedagogy Used for that SLO	Assessment Strategies
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Any Additional Suggestion:
To enhance overall performance, here are some targeted suggestions to help improve students' performance on the quartile concept, especially when working with frequency distribution tables:
Use Graphical Representation: Incorporate cumulative frequency graphs (ogives) to locate quartiles visually.
Work with Real Data: Use everyday data sets (e.g., ages, daily temperatures) to make the concept relatable.

Question No. 3

Question Text

The recorded temperature of a town for 10 days is shown in the given table. The second row shows the difference between daily temperature and mean temperature of the 10 days.

Daily Temperature (X_i)	12	12	14	15	16	17	20	22	22	20
$X_i - \overline{X}$	-5	-5	-3	-2	-1	0	3	5	5	3
$\left(X_i - \overline{X}\right)^2$										

Complete the given table to find the standard deviation and variance in the temperatures.

SLO No.

12.1.3

SLO Text

Calculate variance and standard deviation for ungrouped data by using formulae:

a. variance : $\sigma^2 = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2$ or $\sigma^2 = \frac{\sum \left(x - \bar{x}\right)^2}{n}$

b. standard deviation : $\sigma = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$ or $\sigma = \sqrt{\frac{\sum \left(x - \bar{x}\right)^2}{n}}$

Max Marks

3

Cognitive Level

A

Checking Hints

1 mark for finding values of $\left(X_i - \overline{X}\right)^2$
1 mark for standard deviation
1 mark for finding variance

Overall Performance

Overall, performance on this question was mixed. Some candidates demonstrated a clear understanding of how to calculate squared deviations, variance, and standard deviation

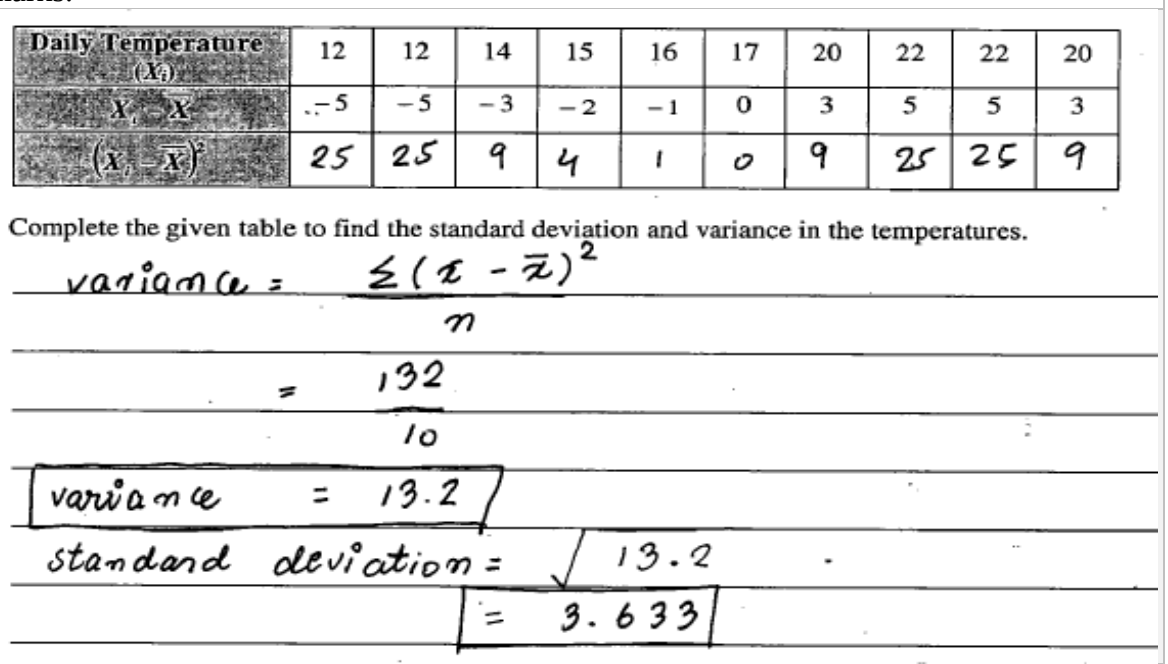
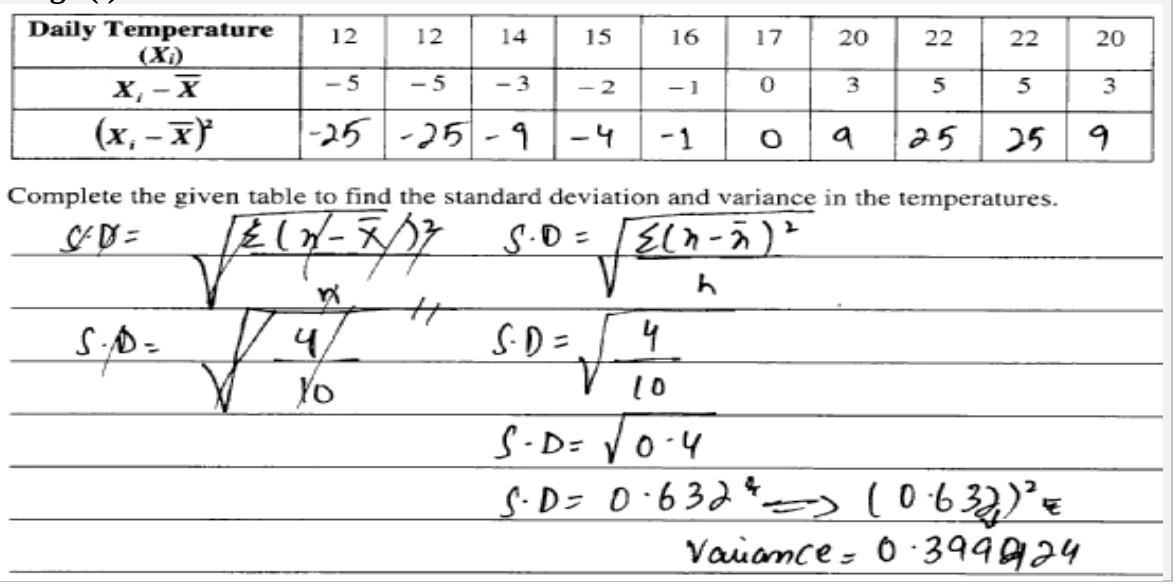
	using the correct formulas. Their work was accurate and well-structured. However, a large number of candidates showed fundamental misunderstandings. Common errors included incorrectly squaring negative deviations, using the wrong formulas, and confusing temperatures with deviations. These mistakes indicate the need for stronger conceptual understanding and more focused practice on these statistical measures.
Description of Better Responses	In *better responses*, candidates correctly used the given mean deviations by squaring each value, finding the total, and accurately substituting into the formula. They completed the steps of dividing by the number of values and taking the square root to find the standard deviation. Their calculations were accurate and followed the correct order, leading to full marks.
Image of Better Response	 Daily Temperature (X_i): 12, 12, 14, 15, 16, 17, 20, 22, 22, 20 $X_i - \overline{X}$: −5, −5, −3, −2, −1, 0, 3, 5, 5, 3 $(X_i - \overline{X})^2$: 25, 25, 9, 4, 1, 0, 9, 25, 25, 9 Complete the given table to find the standard deviation and variance in the temperatures. variance = $\frac{\Sigma(x-\bar{x})^2}{n}$ = $\frac{132}{10}$ variance = 13.2 standard deviation = $\sqrt{13.2}$ = 3.633
Description of Weaker Responses	Many candidates made errors while squaring the deviations, often retaining the negative signs, which led to incorrect values for both variance and standard deviation. Some responses showed a misunderstanding of the method by squaring the temperatures themselves and substituting these values into the standard deviation formula. In other cases, although the correct formulas were used, candidates substituted incorrect values, such as the sum of temperatures instead of the sum of squared deviations, which resulted in further mistakes in the final answers. These errors highlight the need for a clearer understanding of each step in the calculation process.
Images of Weaker Responses	**Image (i)**  Daily Temperature (X_i): 12, 12, 14, 15, 16, 17, 20, 22, 22, 20 $X_i - \overline{X}$: −5, −5, −3, −2, −1, 0, 3, 5, 5, 3 $(x_i - \overline{x})^2$: -25, -25, -9, −4, -1, 0, 9, 25, 25, 9 Complete the given table to find the standard deviation and variance in the temperatures. S.D = $\sqrt{\frac{\Sigma(x-\bar{x})^2}{n}}$ S.D = $\sqrt{\frac{\Sigma(x-\bar{x})^2}{n}}$ S.D = $\sqrt{\frac{4}{10}}$ S.D = $\sqrt{\frac{4}{10}}$ S.D = $\sqrt{0.4}$ S.D = 0.632 ⟹ $(0.632)^2$ Variance = 0.399424

Image (ii)

Daily Temperature (X_i)	12	12	14	15	16	17	20	22	22	20
$X_i - \overline{X}$	−5	−5	−3	−2	−1	0	3	5	5	3
$(X_i - \overline{X})^2$	25	25	9	4	1	0	9	25	25	9

Complete the given table to find the standard deviation and variance in the temperatures.

$SD = \sqrt{\frac{(X-\overline{X})^2}{n}}$

$= \sqrt{\frac{(132-170)^2}{10}}$

$= \sqrt{\frac{(-38)^2}{10}}$

$= \sqrt{144.4}$ $\quad SD = 12.04$ ans

Variance $= \frac{(X-\overline{X})^2}{n}$

$= \frac{(132-170)^2}{10}$

$= \frac{(-38)^2}{10}$

$= 144.4$ ans

Image (iii)

Daily Temperature (X_i)	12	12	14	15	16	17	20	22	22	20
$X_i - \overline{X}$	−5	−5	−3	−2	−1	0	3	5	5	3
$(X_i - \overline{X})^2$	144	144	196	225	256	289	400	484	484	400

Complete the given table to find the standard deviation and variance in the temperatures.

$S.D \rightarrow \sqrt{\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2}$

$\Rightarrow \sqrt{\frac{2797}{10} - \left(\frac{170}{10}\right)^2}$

$= \sqrt{279.7 - (17)^2}$

$= \sqrt{279.7 - 289}$

$= \sqrt{-9.3}$

Suggestions for improvement (Highlight all that apply)

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• Utilise the resource guide for additional materials	• Practical Demonstration	

Any Additional Suggestion:
To enhance overall performance, here are targeted strategies to strengthen students' understanding and application of dispersion measures.

Address Common Errors Proactively:
Use mini quizzes on error patterns that include examples where students mistakenly square negative deviations with the sign, square temperatures instead of deviations, or substitute wrong values into the formula, then review and correct these together as a class.

Encourage Calculator and Spreadsheet Use:
Teach students how to compute variance and standard deviation using calculators or software like Excel or Google Sheets for large data sets to build fluency and help verify manual calculations.

Question No. 4

Question Text

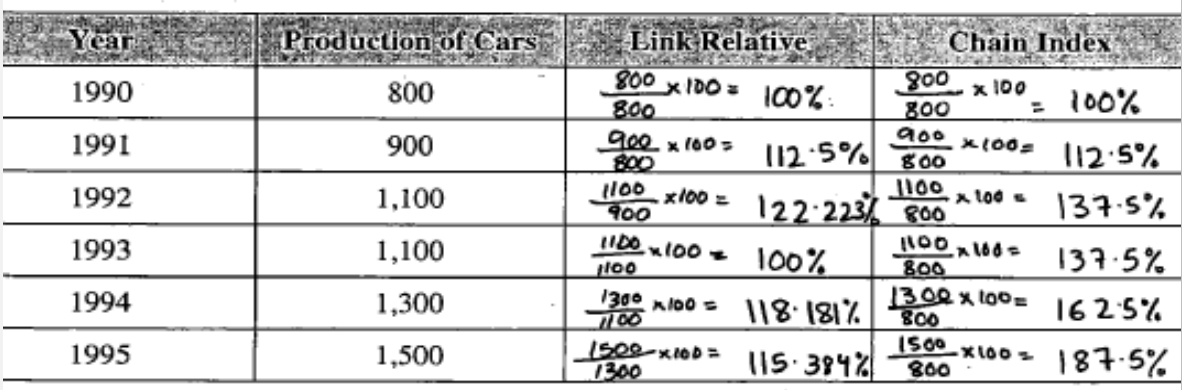
The given table shows the production of cars in a factory is given for the years 1990 to 1995.

Year	Production of Cars	Link Relative	Chain Index
1990	800		
1991	900		
1992	1,100		
1993	1,100		
1994	1,300		
1995	1,500		

Calculate the index number by chain base method using 1990 as a base year.

SLO No.	13.2.1
SLO Text	Calculate index number by fixed base method and chain base method.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for calculating the link relative for three years (2 marks) 1 mark for calculating the chain indices for two years (2 marks)
Overall Performance	More than half of the candidates answered this question correctly, applying both the link relative and chain index methods with a clear understanding of the formulas and calculation steps. The rest showed confusion in selecting and using the correct formulas, leading to errors in their calculations and reducing the accuracy of their responses.
Description of Better Responses	In *better responses*, candidates correctly calculated the link relatives by dividing each value by the one preceding it, showing a sound understanding of the method. For chain indices, they accurately used 1990 as the base year and calculated the percentages accordingly for both required columns. These candidates followed the correct steps and applied the formulas consistently, resulting in accurate and well-structured answers.

Image of Better Response



Year	Production of Cars	Link Relative	Chain Index
1990	800	$\frac{800}{800} \times 100 = 100\%$	$\frac{800}{800} \times 100 = 100\%$
1991	900	$\frac{900}{800} \times 100 = 112.5\%$	$\frac{900}{800} \times 100 = 112.5\%$
1992	1,100	$\frac{1100}{900} \times 100 = 122.22\%$	$\frac{1100}{800} \times 100 = 137.5\%$
1993	1,100	$\frac{1100}{1100} \times 100 = 100\%$	$\frac{1100}{800} \times 100 = 137.5\%$
1994	1,300	$\frac{1300}{1100} \times 100 = 118.181\%$	$\frac{1300}{800} \times 100 = 162.5\%$
1995	1,500	$\frac{1500}{1300} \times 100 = 115.384\%$	$\frac{1500}{800} \times 100 = 187.5\%$

Description of Weaker Responses	Many candidates calculated the chain index using the formula for link relatives, showing confusion between the two methods. Some divided by the total number of cars produced rather than the previous year's value when calculating link relatives, leading to incorrect results. A few candidates left the link relative column blank altogether, while others incorrectly added all the link relative percentages and presented the sum as their final answer. These errors indicate a lack of understanding of the distinction between the two methods and the correct procedures involved.
Images of Weaker Responses	**Image (i)**

Year	Production of Cars	Link Relative	Chain Index
1990	800	800/700×100=11.940298[illegible]	[illegible] 100 %
1991	900	13.43283582	112.5
1992	1,100	16.417 91045	122.2222222
1993	1,100	16.91791045	100%
1994	1,300	19.40298507	118.1818182
1995	1,500	22.3880597	115.3846154

700

Image (ii)

Year	Production of Cars	Link Relative	Chain Index
1990	800		1.875
1991	900		1.125
1992	1,100		1.375
1993	1,100		1
1994	1,300		1.181
1995	1,500		1.153

Calculate the index number by chain base method using 1990 as a base year.

Index 1991 = 900 = 1.125, 1992 = 1,100 = 1.375 | 1995 = 1500 = 1.875
1990 800 1990 800 | 1990 800

Image (iii)

Year	Production of Cars	Link Relative	Chain Index
1990	800		800/800 x100 = 100
1991	900		900/800 x100 = 112.5
1992	1,100		1100/900 x 100 = 122.2
1993	1,100		1100/1100 x 100 = 100
1994	1,300		1300/1100 x100 1.18
1995	1,500		1500/1300 x100 1.15

Calculate the index number by chain base method using 1990 as a base year.

100 + 112.5 + 122.2 + 100 + 1.18 + 1.15

= 436.53 36.53 Percent increase.

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide)	• Story Board • Cause and Effect • Fish and Bone	• Past paper questions • Discussion on E-Marking Notes

• Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

To enhance overall performance, here are targeted strategies to strengthen students' understanding and application of index numbers.

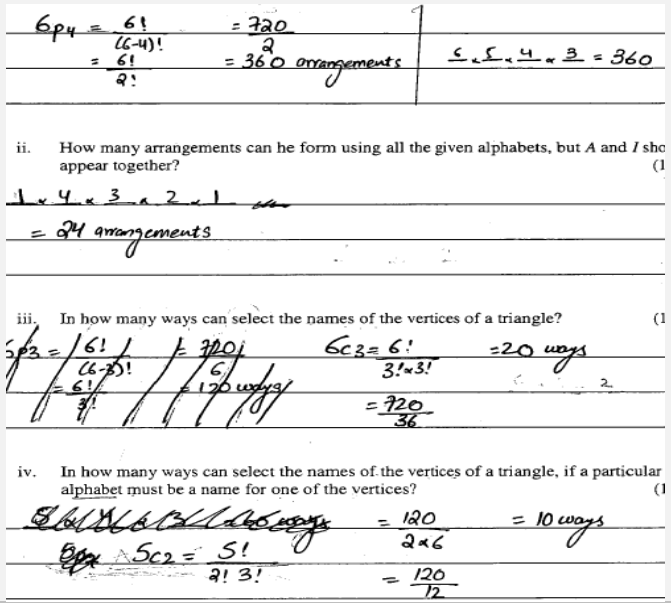
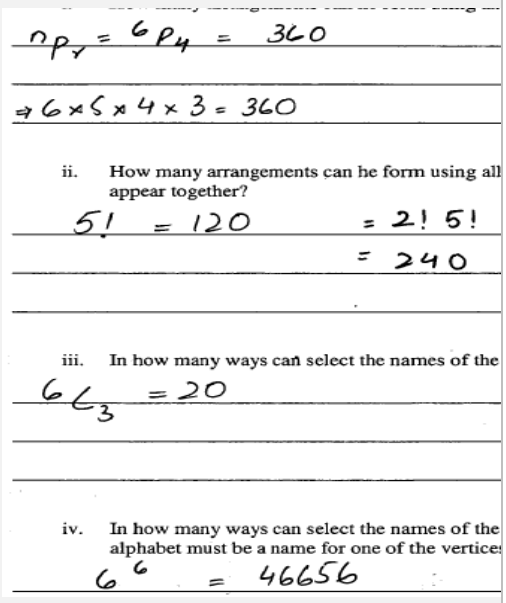
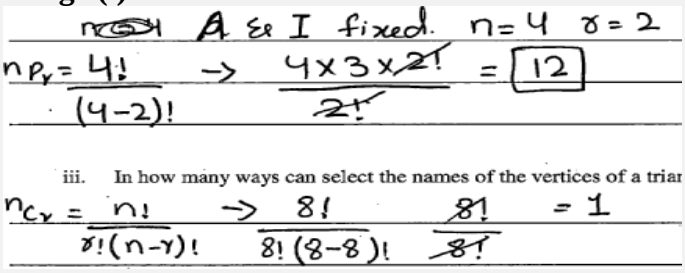
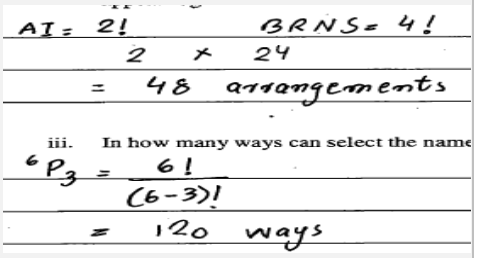
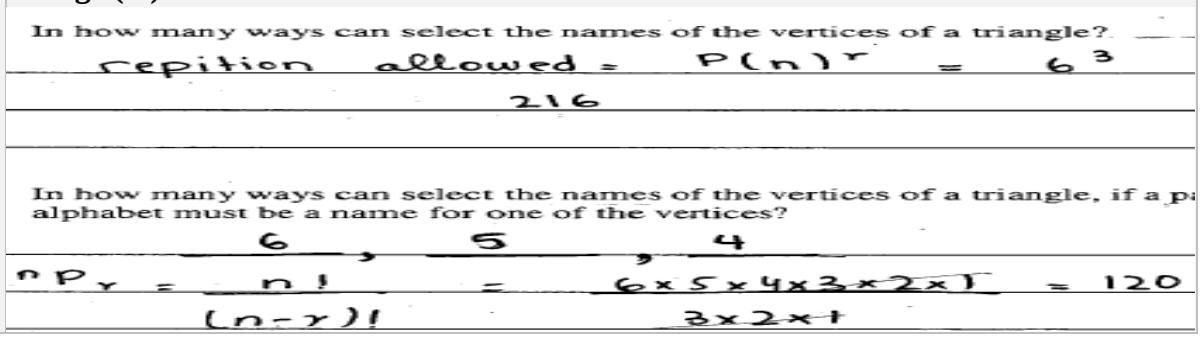
Real-Life Data Projects: Engage students by using real-world data such as car production, fuel prices, or school attendance across different years, having them build tables and calculate link relatives and chain indices to help them connect the concept to actual trends and patterns.

Interactive Digital Spreadsheets: Enhance understanding by using smart boards or tools like Excel/ Google Sheets to input values live, instantly demonstrating how formulas work, which benefits visual learners and provides immediate feedback.

Question No. 5a

Candidates were given the choice to attempt any ONE out of the two questions: 5a and 5b.

Question Text	Akram has a box containing the letters *B*, *R*, *A*, *I*, *N* and *S*. i. How many arrangements can he form using any four alphabets at a time? ii. How many arrangements can he form using all the given alphabets, but *A* and *I* should appear together? iii. In how many ways can select the names of the vertices of a triangle? iv. In how many ways can select the names of the vertices of a triangle, if a particular alphabet must be a name for one of the vertices?
SLO No.	14.3.2 and 14.2.3
SLO Text	Apply the formula of combination to solve related problems. Solve problems related to permutations with and without restrictions.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for each correct part (4 marks)
Overall Performance	Overall, candidates performed well in the first part of the question, demonstrating a basic understanding of the concepts. However, many showed confusion in the remaining parts, mainly due to the incorrect use of permutation and combination formulae. Errors included choosing the wrong formula for the given situation and incorrect substitution of values, which affected the accuracy of their final answers. This suggests the need for more focused practice on when and how to apply these formulae correctly.
Description of Better Responses	In *better responses*, some candidates answered the first three parts of the question correctly. They were able to choose the correct method, permutation or combination, and used the correct values in the formulae. Others gained marks in the first, third, and last parts, where they showed clear working and used factorial expressions properly. These candidates showed they could apply the basic rules of permutations and combinations in the correct way.

Images of Better Responses	**Image (i)**  **Image (ii)** 
Description of Weaker Responses	Most candidates attempted part (i) correctly, showing a good understanding of the basic concept. However, responses to parts (ii), (iii), and (iv) were mixed. Many candidates confused permutations with combinations, often using incorrect forms such as ${}^6C_3$ instead of ${}^6P_3$ or applying the combination formula with incorrect values. These errors suggest uncertainty in distinguishing between situations that require permutations versus combinations. The final part of the question was frequently left unanswered, indicating a lack of confidence or understanding of the concept being tested.
Images of Weaker Responses	**Image (i)**  **Image (ii)**  **Image (iii)** 

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide)	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

• Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	

Any Additional Suggestion:

To enhance overall performance, here are targeted strategies to strengthen students' understanding and application of permutations, combinations, and probabilities.

Real-Life Scenarios Role Play:

To help students grasp the difference between permutations and combinations, engage them in role-playing real-world scenarios. For example, have them form a 3-member team (demonstrating combinations where order doesn't matter) or select 1st, 2nd, and 3rd place winners from 5 students (illustrating permutations where order matters). This hands-on approach allows students to physically experience and internalise the distinction between ordered and unordered selections.

Card Sorting Game:

Reinforce understanding through a card-matching game where students link problem types (permutation or combination), their corresponding formulae, and practical examples. By sorting and matching these elements, students actively practice identifying which concept applies to a given scenario, solidifying their ability to select and apply the correct formula.

Dice or Number Token Experiments:

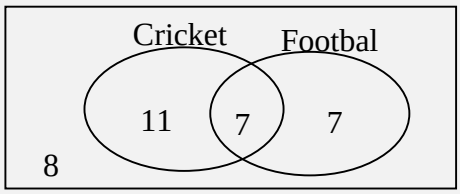
Use dice, number tiles, or playing cards to create interactive experiments. Have students roll the dice or draw tokens, then calculate how many ways they can arrange or select the numbers. Encourage them to list outcomes manually and compare their results with formula-derived answers, bridging the gap between concrete examples and abstract mathematical principles.

Create Your Problem Task:

Deepen conceptual clarity by asking students to design their own permutation or combination problems based on everyday situations like food orders, seating arrangements, or team selections. After crafting their problems, they swap with classmates to solve them, fostering peer learning and reinforcing the practical application of the concepts.

Question No. 5b

Candidates were given the choice to attempt any ONE out of the two questions: 5a and 5b.

Question Text	Faiz asked 33 students whether they liked watching cricket or football matches on TV. The result is shown in the given Venn diagram.  If one student is chosen at random, find the probability that s/he likes watching i. only cricket. ii. cricket or football. iii. cricket and football. iv. either both or neither.
SLO No.	14.4.12
SLO Text	Apply the laws mentioned in SLOs 14.4.5, 14.4.7 and 14.4.10 to solve related problems.

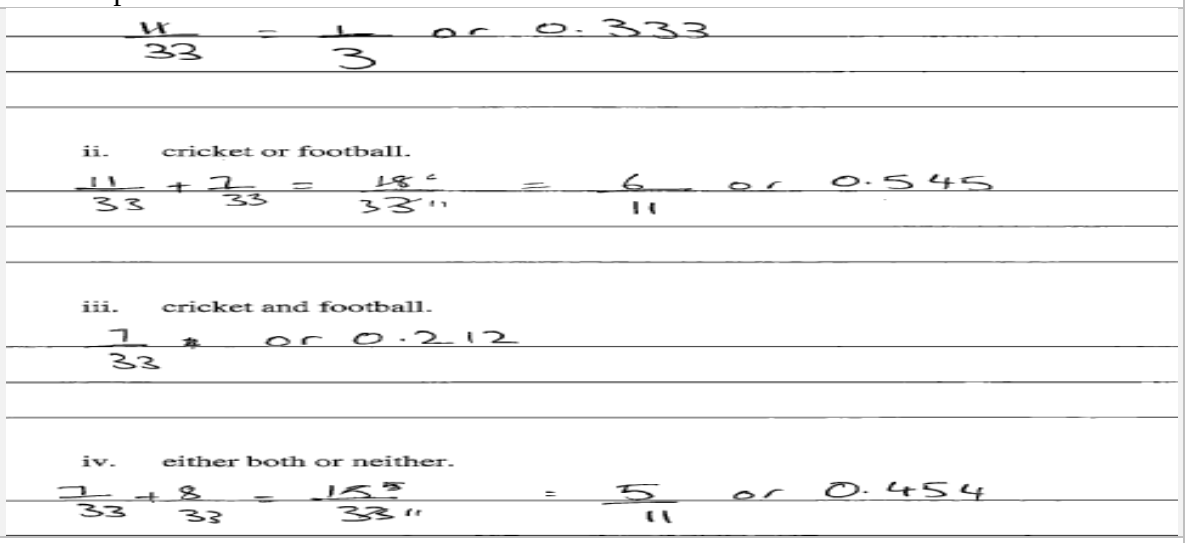
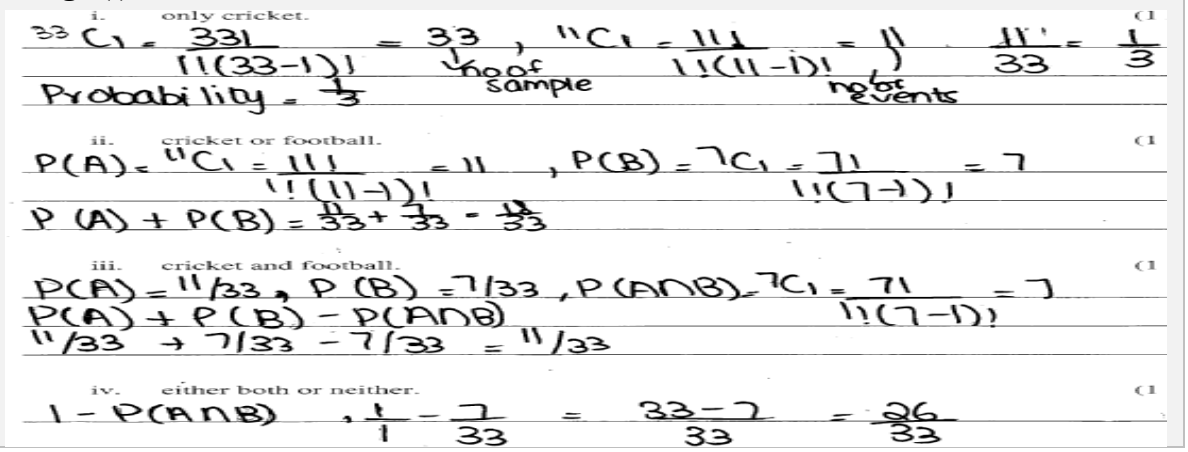
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for each correct part (4 marks)
Overall Performance	Overall, candidates demonstrated a fair understanding of the Venn diagram concept. Many were able to represent and interpret the information clearly within the diagram. However, several responses were affected by errors in applying probability formulas, such as incorrect use of intersections or unions, and by using the wrong total when calculating probabilities. These mistakes reduced the accuracy of the final answers and suggest a need for further practice in combining Venn diagrams with probability calculations.
Description of Better Responses	Candidates correctly used the Venn diagram to identify the number of students who played cricket, football, both or neither. Their responses demonstrated a clear understanding of set representation and accurate interpretation of the data. When calculating probabilities, most candidates correctly divided by the appropriate total, showing a sound grasp of basic probability concepts alongside Venn diagram usage. This reflected a good overall command of the topic.
Image of Better Response	
Description of Weaker Responses	In *weaker responses*, performance varied. Some candidates correctly identified the number of students in each section of the Venn diagram but divided by 25 instead of using the correct total number of students, resulting in incorrect probability values. Others accurately found how many students played each game, but applied the wrong probability formula, such as using addition or multiplication incorrectly, which affected the accuracy of their answers. These errors indicate a need for a better understanding of how to combine Venn diagram interpretation with the correct application of probability formulae.
Images of Weaker Responses	**Image (i)** 

	Image (ii)	Image (iii)
	$P(A) = \frac{n(A)}{n(S)} = \frac{1}{33} = 0.03$ ii. cricket or football. $P(A) = \frac{n(A)}{n(S)} = \frac{1}{11} = 0.09$ $P(B) = \frac{1}{7} = 0.14$ $P(A) + P(B) \Rightarrow 0.09 + 0.14 = 0.23$ iii. cricket and football. $P(A) = \frac{n(A)}{n(S)} = \frac{1}{7} = 0.14$ iv. either both or neither. $P(A) = \frac{n(A)}{n(S)} = \frac{1}{7} = 0.14$ $P(B) = \frac{n(B)}{n(S)} = \frac{1}{8} = 0.125$ $P(A) \times P(B) \Rightarrow 0.14 \times 0.125 = 0.265$	$\frac{11}{25} = 0.44$ ii. cricket or football. $\frac{7}{25} = 0.28$ iii. cricket and football. $\frac{18}{25} = 0.72$ $\frac{7}{25} = 0.28$ iv. either both or neither. 0

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

To enhance overall performance, here are targeted strategies to strengthen students' understanding and performance in Venn diagrams and probability:

Using Simple, Real-Life Contexts:

Begin by introducing Venn diagrams through relatable scenarios familiar to pupils. For instance, analyse choices between extracurricular activities, elective subject selections, or beverage preferences (e.g., "students who drink tea, coffee, both, or neither"). These everyday examples help build conceptual understanding before progressing to abstract numerical problems, demonstrating how Venn diagrams organise real-world information logically.

Practical Activities with Venn Circles:

Create hands-on learning experiences using hula hoops or rope circles on the floor to physically represent sets. Have pupils place themselves or labelled cards in the correct regions (such as "Football only", "Rugby only", "Both sports", or "Neither"). This kinaesthetic approach makes abstract set relationships tangible, reinforcing understanding through physical participation and visualisation.

Structured Table-to-Diagram Practice

Develop pupils' fluency in interpreting and creating Venn diagrams through scaffolded exercises. Provide frequency tables and challenge learners to convert these into accurate Venn diagrams, then reverse the activity by having them extract data from completed diagrams back into tables. This deliberate practice builds essential skills in translating between different mathematical representations.

Interactive Games and Quizzes:

Incorporate engaging digital tools like Kahoot or Quizizz for lively quizzes testing identification of Venn regions and probability calculations. Alternatively, create card-matching activities where pupils pair data sets with corresponding diagrams and probability notation (such as "P(A∪B)"). These interactive methods promote active recall while making revision collaborative and enjoyable.

Question No. 6

Question Text

The data shown in the given table represents the advertisement cost (in thousand rupees) and sales (in hundred thousand rupees) in a company. Using the given information, calculate the coefficient of correlation.

Advertisement Cost (in Thousand Rupees, x)	Sales (in Hundred Thousands Rupees, y)	x^2	y^2	xy
40	38	1600	1444	1520
39	43	1521	1849	1677
45	38	2025	1444	1710
49	45	2401	2025	2205
48	49	2304	2401	2352
50	47	2500	2209	2350
47	48	2209	2304	2256
44	44	1936	1936	1936
46	45	2116	2025	2070
48	48	2304	2304	2304

SLO No.

15.2.3

SLO Text

Calculate the coefficient of correlation by using the formula.

$$r = \frac{n\sum XY - (\sum X)(\sum Y)}{\sqrt{\left[n\sum X^2 - (\sum X)^2\right]\left[n\sum Y^2 - (\sum Y)^2\right]}}$$

Max Marks

3

Cognitive Level

U

Checking Hints

1 mark for finding the sums
1 mark for extracting the required values correctly
1 mark for calculating the correlation coefficient

Overall Performance

Overall, candidates showed a fair understanding of the concept. However, many responses were affected by mistakes in substitution and simplification of values within the formulas. These errors often led to incorrect final answers, despite an otherwise correct approach. To improve performance, candidates would benefit from more practice in accurately applying and simplifying mathematical formulae step by step.

Description of Better Responses

In *better responses*, candidates correctly calculated the total sums for each variable and used them in the formula for the coefficient of correlation. They substituted the right values into the formula, followed the steps carefully, and simplified the expression properly. Their methodical approach showed a good understanding of the process and led to the correct value of the correlation coefficient.

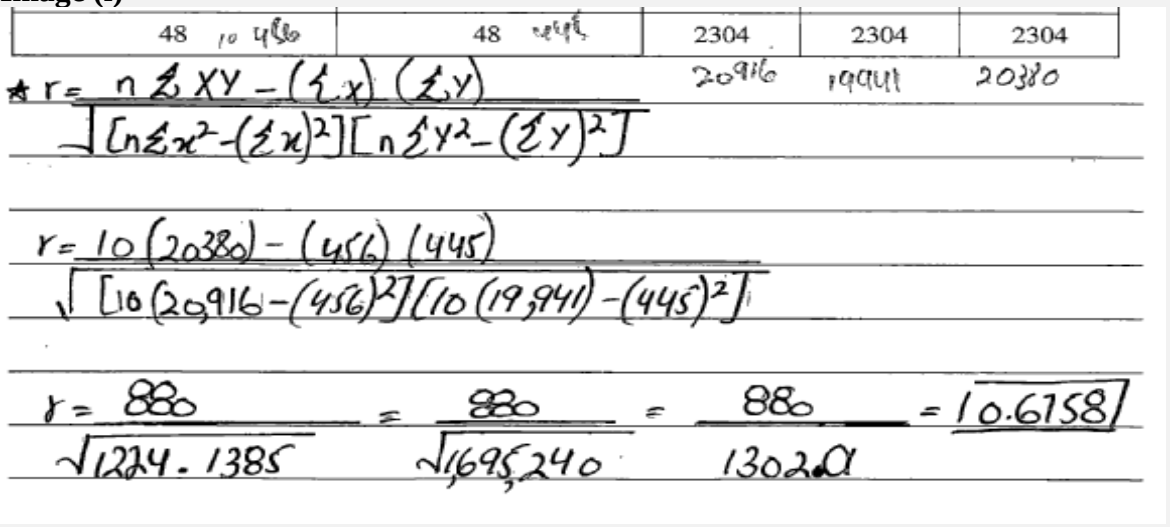
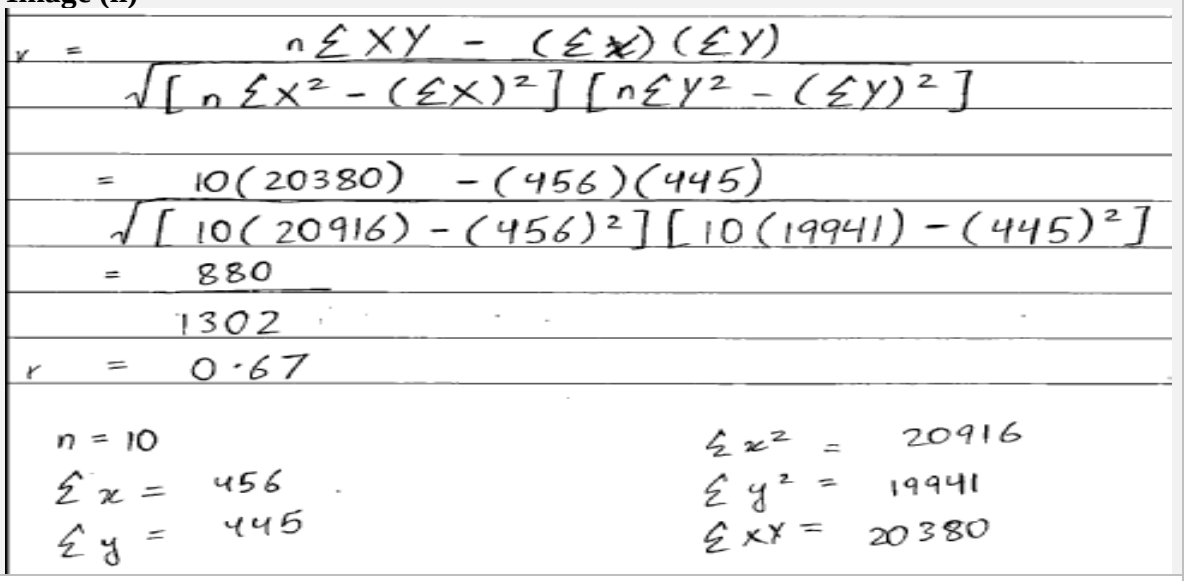
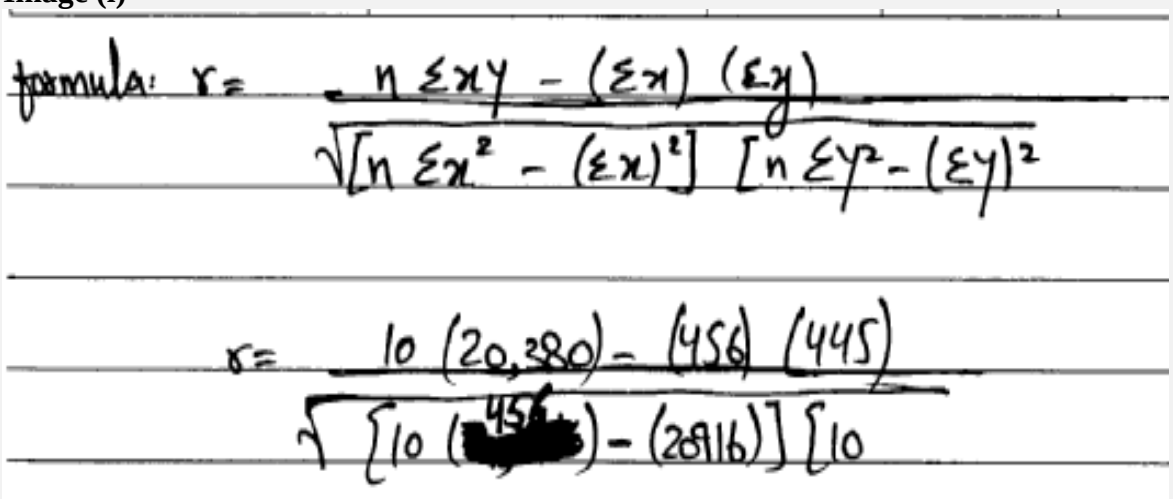
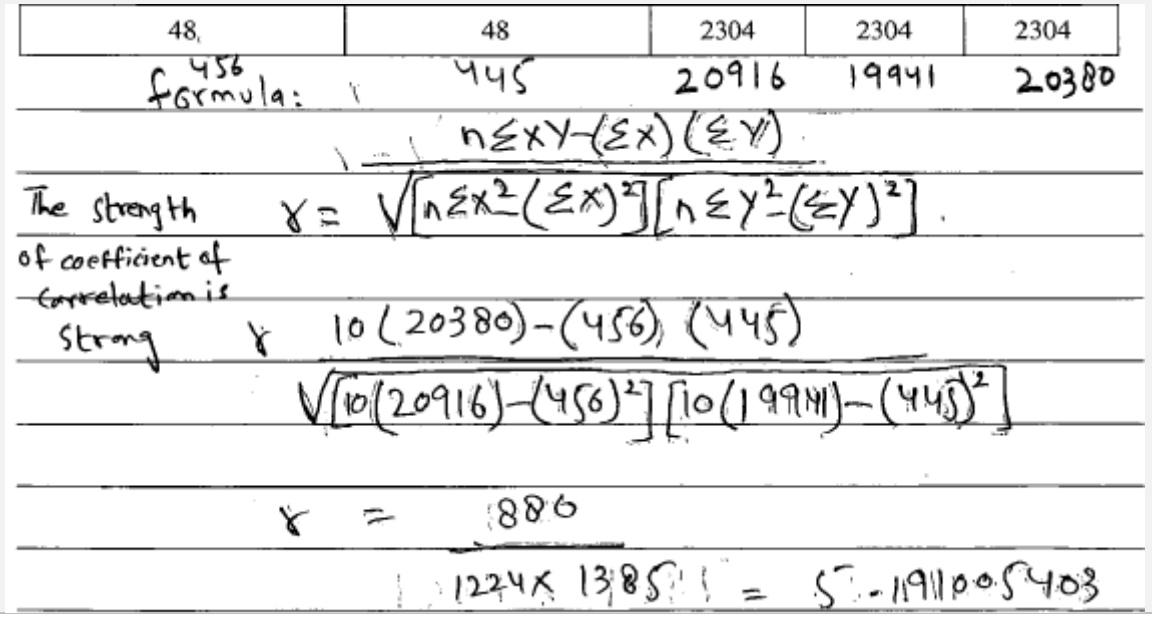
Images of Better Responses	**Image (i)**  48 48 2304 2304 2304 20916 19941 20380 $r = \frac{n\Sigma XY - (\Sigma x)(\Sigma y)}{\sqrt{[n\Sigma x^2 - (\Sigma x)^2][n\Sigma y^2 - (\Sigma y)^2]}}$ $r = \frac{10(20380) - (456)(445)}{\sqrt{[10(20,916) - (456)^2][10(19,941) - (445)^2]}}$ $r = \frac{880}{\sqrt{1224.1385}} = \frac{880}{\sqrt{1,695,240}} = \frac{880}{1302.01} = 0.6758$ **Image (ii)**  $r = \frac{n\Sigma XY - (\Sigma x)(\Sigma y)}{\sqrt{[n\Sigma x^2 - (\Sigma x)^2][n\Sigma y^2 - (\Sigma y)^2]}}$ $= \frac{10(20380) - (456)(445)}{\sqrt{[10(20916) - (456)^2][10(19941) - (445)^2]}}$ $= \frac{880}{1302}$ $r = 0.67$ $n = 10$ $\Sigma x = 456$ $\Sigma y = 445$ $\Sigma x^2 = 20916$ $\Sigma y^2 = 19941$ $\Sigma xy = 20380$
Description of Weaker Responses	In *weaker responses*, many candidates were able to calculate the correct totals for each variable. However, they made mistakes when substituting these values into the correlation coefficient formula. Some candidates used the wrong values in place of the variables, while others who substituted correctly made errors when simplifying the expression. As a result, they were unable to reach the correct value of the correlation coefficient.
Images of Weaker Responses	**Image (i)**  formula: $r = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{\sqrt{[n\Sigma x^2 - (\Sigma x)^2][n\Sigma y^2 - (\Sigma y)^2}}$ $r = \frac{10(20,380) - (456)(445)}{\sqrt{[10([redacted]) - (20916)][10}}$

Image (ii)



48	48	2304	2304	2304

formula: $\sum x = 456$, $\sum y = 445$, $\sum xy = 20916$, $\sum x^2 = 19941$, $\sum y^2 = 20380$

$$r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

The strength of coefficient of correlation is strong

$$r = \frac{10(20916) - (456)(445)}{\sqrt{[10(20916) - (456)^2][10(19941) - (445)^2]}}$$

$$r = \frac{886}{1224 \times 1385} = 5.19110054 03$$

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	**Preferred Pedagogy** Used for this SLO**	**Assessment Strategies**
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

By implementing these strategies, educators can enhance students' understanding of the concept of correlation.

Effective Calculator Skills:

Teach pupils to properly utilise scientific calculators for squaring values, multiplication, and simplifying expressions to reduce computational errors. Demonstrate how to use the memory function for multi-step calculations and emphasise checking each entry. Regular practice with teacher supervision helps build both competence and confidence when handling statistical operations.

Spreadsheet Competency Development:

Guide pupils through using spreadsheet software (Excel or Google Sheets) for statistical analysis by: Properly entering data sets, employing SUM functions for totals, allows learners to concentrate on methodological accuracy while the software handles complex arithmetic, developing both their statistical understanding and digital skills.

Annexure A: Pedagogies Used for Teaching the SLOs

Pedagogy: Storyboard
Description: A visual pedagogy that uses a series of illustrated panels to present a narrative, encouraging creativity and critical thinking. It helps learners organise ideas, sequence events, and comprehend complex concepts through storytelling.
Example: In a Literature class, students are tasked with creating storyboards to visually retell a novel. They draw key scenes, write captions, and present their stories to the class, enhancing their reading comprehension and fostering their imagination.

Pedagogy: Cause and Effect
Description: This pedagogy explores the relationships between actions and consequences. By analysing cause-and-effect relationships, learners develop a deeper understanding of how events are interconnected and how one action can lead to various outcomes.
Example: In a History class, students study the causes and effects of the Industrial Revolution. They research and discuss how technological advancements in manufacturing led to significant societal changes, such as urbanisation and labour reform movements.

Pedagogy: Fish and Bone
Description: A method that breaks down complex topics into main ideas (the fish) and supporting details (the bones). This visual approach enhances comprehension by highlighting essential concepts and their relevant explanations.
Example: During a Biology class on human anatomy, the teacher uses the fish and bone technique to teach about the human skeletal system. Teacher presents the main components of the human skeleton (fish) and elaborates on each bone's structure and function (bones).

Pedagogy: Concept Mapping
Description: An effective way to visually represent relationships between ideas. Learners create diagrams connecting key concepts, aiding in understanding the overall structure of a subject and fostering retention.
Example: In a Psychology assignment, students use concept mapping to explore the various theories of personality. They interlink different theories, such as Freud's psychoanalysis, Jung's analytical psychology, and Bandura's social-cognitive theory, to see how they relate to each other.

Pedagogy: Audio Visual Resources
Description: Incorporating multimedia elements like videos, images, and audio into lessons. This approach caters to different learning styles, making educational content more engaging and memorable.
Example: In a General Science class, the teacher uses a documentary-style video to teach about the solar system. The video includes stunning visual animations of the planets, interviews with astronomers, and background music, enhancing students' interest and understanding of space.

Pedagogy: Think, Pair, and Share
Description: A collaborative learning technique where students ponder a question or problem individually, then discuss their thoughts in pairs or small groups before sharing with the entire class. It fosters active participation, communication skills, and diverse perspectives.
Example: In a Literature in English class, the teacher poses a thought-provoking question about a novel's moral dilemma. Students first reflect individually, then pair up to exchange their opinions, and finally participate in a lively class discussion to explore different viewpoints.

Pedagogy: Questioning Technique (Socratic Approach)
Description: Based on Socratic dialogue, this method stimulates critical thinking by posing thought-provoking questions. It encourages learners to explore ideas, justify their reasoning, and discover knowledge through a process of inquiry.
Example: In an Ethics class, the instructor uses the Socratic approach to lead a discussion on the meaning of justice. By asking a series of probing questions, the students engage in a deeper exploration of ethical principles and societal values.

Pedagogy: Practical Demonstration
Description: A hands-on approach where learners observe real-life applications of theories or skills. Practical demonstrations enhance comprehension, skill acquisition, and problem-solving abilities by bridging theoretical concepts with real-world scenarios.
Example: In a Food and Nutrition class, the instructor demonstrates the proper technique for filleting a fish. Students observe and then practice the skill themselves, learning the practical application of knife skills and culinary precision.

(**Note:** The examples provided in this annexure serve as illustrations of various pedagogies. It is important to understand that these pedagogies are versatile and can be applied across subjects in numerous ways. Feel free to adapt and explore these techniques creatively to enhance learning outcomes in your specific context.)

Acknowledgements

The Aga Khan University Examination Board (AKU-EB) acknowledges with gratitude the invaluable contributions of all the dedicated individuals who have played a pivotal role in the development of the Business Statistics HSSC-II E-Marking Notes.

We extend our sincere appreciation to Ms Shahneel Jawaid and Mr Muhammad Kasshaf Shaikh, Specialists in Mathematics at AKU-EB, for taking subject lead during the entire process of e-marking.

We particularly thank to Ms Saba Naz, Habib Girls' High School, Karachi, for evaluating each question's performance, delineating strengths and weaknesses in candidates' responses, and highlighting instructional approaches along with recommendations for better performance.

Additionally, we express our gratitude to the esteemed team of reviewers for their constructive feedback on overall performance, better and weaker responses, and validating teaching pedagogies, along with suggestions for improvement.

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- Dr Naveed Yousuf, CEO, AKU-EB
- Zain Muluk, Manager, Examination Development, AKU-EB
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